

SATRA

Spaceborne Automated Thermal Radiative Anomaly Classification System

Hackathon: Smart India Hackathon (SIH)	Problem Statement: PS 26162	Model Version: v2.0.0-scientific
Classification: 4-Class Radiative AI	Telemetry: 100% Real NASA FIRMS	Status: Production & Demo Ready

EXECUTIVE SUMMARY: SATRA is an operational, AI-powered intelligence platform engineered to bridge the gap between spaceborne thermal remote sensing and ground-truth verification. Engineered specifically for **Problem Statement 26162**, SATRA addresses a critical national challenge: rapidly identifying, classifying, and verifying industrial and wildfire thermal anomalies across India and globally using real NASA spaceborne sensors (VIIRS NOAA-20, NOAA-21, S-NPP and MODIS Terra/Aqua). SATRA operates exclusively on real satellite telemetry—**zero mock or simulated data**—and features an integrated pipeline that ingests raw spaceborne thermal observations, computes multi-band physical features, executes an ensemble AI classifier, queries OpenStreetMap/Overpass GIS physical infrastructure layers, and presents actionable intelligence through 3D geospatial globes and local inspection interfaces.

1. End-to-End System Architecture & Pipeline

SATRA's architecture is composed of four loosely coupled, robust tiers designed for deterministic execution, sub-second inference latency, and high-throughput real-time telemetry processing.

Pipeline Tier	Technical Components	Role & Operational Responsibility
1. Ingestion Layer	FastAPI, NASA FIRMS API, CSV/GeoJSON Parsers	Authenticates against NASA FIRMS, fetches live spaceborne thermal telemetry (VIIRS/MODIS), and parses external uploads.
2. Processing & AI Engine	Scikit-Learn, NumPy, Joblib (Model v2.0.0)	Calculates dual-band temperature differentials (T4-T31), Fire Radiative Power (MW), day/night contrast, and executes 4-class classifier.
3. GIS & Evidence Layer	OpenStreetMap Overpass API, Haversine Engine	Queries physical ground infrastructure within 1-5 km (refineries, flares, industrial zones) to supply corroborative physical evidence.
4. Visualization & Triage	React 19, Vite, Three.js 3D Globe, Leaflet	Renders space orbit down to meters, FRP dynamic heat halos, telemetry popups, triage queue, and RFC-4180 CSV exports.

2. Core Modules & Technological Specifications

2.1 High-Performance Backend & Telemetry Ingestion (FastAPI)

- **Asynchronous REST Backend:** Built with FastAPI running on Python 3.10+, featuring self-documenting OpenAPI schemas, dependency injection, and CORS-secured client communication.
- **NASA FIRMS Ingestion:** Server-side API key management via NASA_FIRMS_MAP_KEY. Fetches active fires from VIIRS NOAA-20, NOAA-21, S-NPP (375m spatial resolution) and MODIS (1km resolution).
- **Strict Data Provenance:** Each ingestion event tags records with explicit origins (REAL_FIRMS, USER_UPLOADED, PROTOTYPE_LABELLED), ensuring verifiable auditability.

2.2 Artificial Intelligence & Physics-Informed Classification

- **Canonical 4-Class Taxonomy:**
 1. **Industrial Fire:** Active refinery flares, petrochemical furnaces, chemical plants, steel mills.
 2. **Forest Fire:** Wildfire fronts, forest canopy blazes, biomass burning, and vegetation anomalies.
 3. **Persistent Thermal Source:** Stationary industrial infrastructure, persistent flares, and continuous heat emitters.
 4. **Other:** Controlled agricultural residue burning, low-radiance anomalies, non-industrial signatures.
- **Feature Vector:** Fire Radiative Power (MW), Brightness Temperature (T4), Dual-band Temperature Differential ($\Delta T = T4 - T31$), Day/Night flag, and local spatial cluster density.
- **Uncertainty Quantification:** Predictions with classification confidence < 60% are flagged as LOW_CONFIDENCE_REVIEW, routing to human operators to eliminate false positives in critical operations.

2.3 GIS Ground Evidence & Spatial Verification Engine

- **Ground Truth Philosophy:** GIS context serves as corroborative physical evidence—it enhances operator awareness but never replaces physical ground verification.
- **Overpass API Dynamic Querying:** Automatically searches for tagged industrial infrastructure (man_made=flare, landuse=industrial, petrochemical) within a 1 to 5 km radius.
- **6-Layer Cartography:** Supports ESRI World Imagery, OpenStreetMap Standard, Dark Canvas, Topographic Terrain, Street Map, and Telemetry Heatmaps.

2.4 Interactive Frontend & Operational Triage (React + Vite)

- **3D Earth Intelligence:** Three.js interactive 3D Globe with correct geographic orientation (North aligned UP). Camera seamlessly interpolates from orbital space view down to ground coordinates.
- **Operational Alert Lifecycle:** Strict state machine transitions: REQUIRES_VERIFICATION → UNDER_INVESTIGATION → CONFIRMED / DISMISSED, with auditable operator notes.
- **Exporting & Analytics:** Real-time dynamic aggregation and RFC-4180 compliant CSV export.

3. Zero-Mock Real Data Guarantee

DATA INTEGRITY COMMITMENT: To ensure the highest scientific standards for SIH 2026, SATRA underwent an exhaustive repository audit:

1. **Database Purging:** All synthetic/sample mock records were eradicated from thermal_detections.db. Runtime observations stem exclusively from live NASA FIRMS feeds or verified user uploads.
2. **Dynamic Live KPIs:** Removed all static hardcoded percentages and fake UI trends. Every metric is computed dynamically from real database observations.
3. **Transparent Empty States:** In regions or timeframes devoid of satellite passes, the system displays honest empty indicators ('No real satellite observations available') rather than fabricated demo placeholders.

4. Comprehensive Verification & Testing Metrics

The system has been validated through exhaustive automated test suites covering backend APIs, AI inference, data ingestion, and frontend compilation with 100% passing rates.

Automated Test Suite	Scope & Subsystem Tested	Result / Benchmark
test_api_detections	CRUD operations, filtering, pagination, geometry	8 / 8 Passed (100%)
test_api_alerts	Lifecycle state machine & verification notes	5 / 5 Passed (100%)
test_api_inference	Real-time ML inference & database persistence	4 / 4 Passed (100%)
test_api_satellite_firms	NASA FIRMS live client, key auth & rate limits	7 / 7 Passed (100%)
test_scientific_validation	Feature physics, threshold bounds, math constraints	9 / 9 Passed (100%)
test_cleaning_pipeline	Data cleansing, duplicate removal, schema validation	8 / 8 Passed (100%)
Full Backend Pytest Suite	25 test suites across all core modules	145 / 145 Passed in 8.83s
Vite Production Build	Frontend React bundle compilation & asset bundling	Passed (0 errors, <700ms)

5. Deployment, Local Access & Operational Execution

Both frontend and backend services are active and running in the current workspace:

Service Component	Local URL Endpoint	Functionality
Frontend Web App	http://localhost:5173	Vite + React Dashboard, 3D Globe & Map View
Backend REST API	http://localhost:8000	FastAPI Base Root & Health Endpoints
Swagger UI Docs	http://localhost:8000/docs	Interactive API Documentation & Testing
ReDoc Specification	http://localhost:8000/redoc	Alternative OpenAPI Schema Documentation

6. Conclusion & Hackathon Readiness Sign-Off

SATRA delivers a complete, production-grade intelligence system for remote sensing thermal anomaly classification. By fusing NASA spaceborne telemetry with physics-informed AI, real Overpass GIS infrastructure verification, and human-in-the-loop triage workflows—backed by 100% real runtime data and 145 passing automated tests—SATRA provides disaster response agencies, environmental monitors, and industrial auditors with immediate, verifiable, and mission-critical situational awareness.